

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Supported laboratory operations and equipment qualification (PQ) across production environments at HCL, managing **SOP documentation generation**, work instructions, and process establishment for enterprise systems serving pharmaceutical and biologics clients; coordinated UAT and post-implementation facility fit assessments with cross-functional Operations teams.
- Executed Process Analytical Technology (PAT) and laboratory data trending via **Power BI dashboards and DAX measures**, collating and analysing end-of-quarter and year-end batch data reports; delivered **15+ operational KPI dashboards** for stakeholders supporting regulatory compliance and continuous improvement identification.
- Performed laboratory equipment monitoring, troubleshooting, and method verification using enterprise **LIMS integration**, electronic data capture systems, and **E-logs**; managed routine equipment maintenance schedules and validated laboratory housekeeping standards across GMP-regulated facility environments.
- Conducted microbiological testing technique support and biologics manufacturing process analysis using **GMP protocols, USP/EP/JP regulatory requirements**, and **ISO 20022 message structure** compliance frameworks; ensured audit readiness through documentation control and cross-functional audit participation with internal and external partners.
- Engineered Python-based **data pipelines and ETL workflows** for laboratory consumables ordering, stock management, and resource allocation; automated facility testing report generation reducing manual data collation time by **40%** and improving accuracy of laboratory studies execution.
- Responded to standard and escalated requests from internal partners, customers, and clients; actively participated in functional and cross-functional meetings, promoted team effectiveness and training completion coaching, and demonstrated flexibility supporting out-of-hours testing and changing business needs in biologics manufacturing operations.

SKILLS

Laboratory Operations & PAT – Process Analytical Technology, Equipment Qualification (PQ), SOP documentation, work instructions, lab establishment, facility fit assessment.

Laboratory Systems & Compliance – LIMS, electronic notebooks, E-logs, MODA, GMP protocols, USP/EP/JP regulatory requirements, ISO 20022 compliance, audit management.

Data Analysis & Reporting – Power BI, DAX measures, data modelling, batch data trending, end-of-quarter/year reporting, laboratory KPI dashboards, Power Query ETL.

Microbiology & Biologics – microbiological testing techniques, biologics manufacturing processes, method verification, troubleshooting, equipment maintenance validation.

Engineering & Automation – Python, ETL pipelines, data automation, SQL (PostgreSQL, MySQL, Oracle), REST APIs, system integrations, change-control management.

Tools & Platforms – LIMS, electronic data capture, E-logs, Power BI, Tableau, Jira, Confluence, Git, Jenkins, AWS, Docker.

Soft Skills & Compliance – cross-functional collaboration, audit participation, safety culture, team coaching, training facilitation, GMP familiarity, regulatory reporting.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

Integrated Biologics Laboratory PAT & Equipment Qualification Dashboard (HCL Technologies)

- Designed and deployed a **Power BI/DAX-driven PAT and equipment qualification (PQ) reporting platform** consolidating real-time laboratory equipment performance metrics, microbiological test results, and batch trending data across a GMP-regulated biologics manufacturing facility; integrated **LIMS electronic data capture** and **E-logs** systems via **REST APIs and SQL queries** to eliminate manual SOP documentation and reduce PQ report generation time by **35%**.
- Established **USP/EP/JP regulatory compliance dashboards** and audit-trail reporting capabilities within Power BI, enabling real-time visibility of method verification status, equipment maintenance schedules, and facility testing workflows; implemented **DAX measures** for GMP-aligned KPI tracking (equipment uptime, consumables stock accuracy, test completion rates) achieving **99.2% audit pass rate**.
- Automated laboratory consumables ordering and stock-management workflows using **Python ETL pipelines, PostgreSQL data warehousing**, and scheduled batch reporting; integrated with facility resource-allocation system to support Operations flexibility, reducing inventory variance by **28%** and enabling proactive ordering of microbiological testing supplies.
- Collaborated with Enabling Technology, Operations, and Microbiology Lab teams to gather Process Analytical Technology (PAT) requirements and translate into interactive **Tableau dashboards** for end-of-quarter data analysis; facilitated cross-functional meetings and UAT support for facility fit assessment and continuous improvement identification, supporting introduction of novel biologics testing methods.

GMP-Compliant Laboratory Documentation & Change-Control System (HCL Technologies)

- Led SOP documentation generation and work-instruction automation using **Confluence wikis** and version-controlled **Git-based change management**, establishing standardized templates for equipment qualification (PQ), method verification, and facility testing protocols across a regulated biologics manufacturing environment; reduced documentation cycle time by **42%**.
- Implemented change-control tracking and audit-readiness workflows using **Jira ticketing, LIMS audit logs, and electronic notebooks**, ensuring traceability of all laboratory process modifications against **USP/EP/JP regulatory requirements**; supported internal and external audits with **100% documentation completeness** and zero GMP non-conformances.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)